BRAC UNIVERSITY

Department of Electrical and Electronic Engineering
School of Engineering

INTRODUCTION TO COMMUNICATION ENGINEERING

Lecture 02: Amplitude Modulations and Demodulations

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[Date]

EEE341

INTRODUCTION

Modulation often refers to a process that moves the message signal into a specific frequency band that is dictated by the physical channel (e.g., voiceband telephone modems).

Modulation provides a number of **advantages,** including

- ease of RF transmission and
- frequency division multiplexing.

Modulations can be

- **analog or**
- **digital.**

Though traditional communication systems such as AM/FM radios are based on analog modulations, more recent systems such as 2G and 3G cellphones, HDTV, and DSL are all digital.

In this lecture, we will focus on the **classic analog modulations**:

- **Amplitude modulation** and
- **angle modulation.**

Note that it is important to distinguish between communication

- systems that **do not use** modulation - **baseband communications** and
- systems that **use** modulation - **carrier communications.**

BASEBAND VERSUS CARRIER COMMUNICATIONS

The term **baseband** is used to designate the frequency band of the original message signal from the *source or the input transducer* (see **Fig.1.2**).

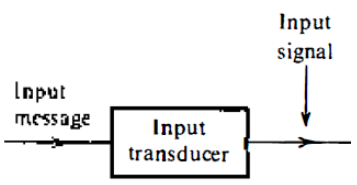


Fig.2.1: Parts of Fig.1.2.

- In telephony, the **baseband is the audio band** (band of voice signals) of 0 to 3.5 kHz.
- For digital data or pulse code modulation (**PCM**) that uses bipolar signaling at a rate of R_b pulses per second, the baseband is approximately 0 to R_b Hz.

Baseband Communications

In baseband communication, **message signals are directly transmitted without any modification.**

Because most baseband signals such as audio and video contain significant **low-frequency content**, they cannot be effectively transmitted over radio (wireless) links.

Instead, **dedicated user channels** such as twisted pairs of copper wires and coaxial cables are assigned to each user for long-distance communications.

Because baseband signals have **overlapping bands**, they would interfere severely if sharing a common channel.

Thus, baseband communications leave much of the channel spectrum unused.

Carrier Modulations

Communication that **uses modulation to shift the frequency spectrum of a signal** (baseband) is known as **carrier communication.**

- In terms of **analog modulation**, one of the basic parameters (amplitude, frequency, or phase) of a **sinusoidal carrier** of high-frequency f_c Hz (or $W_c = 2nf_c$ rad/s) is varied linearly with the baseband signal $m(t)$.

- This results in amplitude modulation (AM), frequency modulation (FM), or phase modulation (PM), respectively.

- **Amplitude modulation** is **linear**, while the latter two types of carrier modulation are **similar and nonlinear**, often known collectively as **angle modulation.**

An important comment about **pulse-modulated signals**

[pulse amplitude modulation (PAM),
pulse width modulation (PWM),
pulse position modulation (PPM),
pulse code modulation (PCM),
and delta modulation (DM)]

is as **follows.**

- Despite the term **modulation,** these signals are **baseband digital** signals.
- "**Modulation**" is used here **not in the sense of frequency or band shifting**.
- Rather, in these cases, it is in fact **describing digital pulse coding schemes** used **to represent the original analog signals**.
- In other words, the analog message signal is modulating the parameters of **a digital pulse train (not another analog high-frequency carrier signal** as in the case of Amplitude or angle modulation**)**.
- **These signals can still modulate a carrier** in order to shift their spectra.

Amplitude Modulations and Angle Modulations

We denote as $m(t)$ the source message signal that is to be transmitted by the sender to its receivers. Let the frequency response of $m(t)$ is moved to a new frequency band centered at f_c Hz (carrier frequency),

Consider a sinusoidal signal (carrier signal)

$$s(t) = A(t)\cos\left[\omega_c t + \phi(t)\right]$$

There are **three** variables in the above sinusoid: amplitude, (instantaneous) frequency, and phase.

Indeed, the **message signal** $m(t)$ can be used to **modulate** any one of these three parameters to allow $s(t)$ to carry the information from the transmitter to the receiver:

Amplitude $A(t)$ linearly varies with $m(t)$	$\Longleftrightarrow$	amplitude modulation
Frequency linearly varies with $m(t)$	$\Longleftrightarrow$	frequency modulation
Phase $\phi(t)$ linearly varies with $m(t)$	$\Longleftrightarrow$	phase modulation

These are known, respectively, as amplitude modulation, frequency modulation, and phase modulation.

DOUBLE-SIDEBAND AMPLITUDE MODULATION

Amplitude modulation is characterized by an information-bearing ***carrier*** amplitude $A(t)$ that is a linear function of the baseband (message) signal $m(t)$.

At the same time, the carrier frequency ω_c and the phase θ_c remain constant.

We can assume $\theta_c = 0$ without loss of generality.

If the carrier amplitude A is made directly proportional to the modulating signal $m(t)$, then **modulated signal** is $m(t)\cos\omega_c t$ (Fig. 4.1).

This type of modulation simply shifts the spectrum of $m(t)$ to the carrier frequency (Fig. 4.1 a). Thus, if

$$m(t) \Longleftrightarrow M(f)$$

Then

$$m(t)\cos 2\pi f_c t \Longleftrightarrow \tfrac{1}{2}[M(f+f_c) + M(f-f_c)] \qquad (4.1)$$

Recall that $M(f-f_c)$ is M (f) shifted to the right by f_c, and $M(f+f_c)$ is $M(f)$ shifted to the left by f_c.

- Thus, the process of modulation shifts the spectrum of the modulating signal **to the left and to the right by f_c.**
- If the bandwidth of $m(t)$ is B Hz, then, as seen from Fig. 4.lc, the modulated signal now has a bandwidth of $2B$ Hz.
- The modulated signal spectrum centered at $\pm f_c$ (or $\pm w_c$, in rad/s) **consists of two parts**:
  - a portion that lies outside $\pm f_c$, known as the **upper sideband (USB)**, and
  - a portion that lies inside $\pm f_c$, known as the **lower sideband (LSB).**
- From Fig. 4.lc, unless the message signal ***M*(f) has an impulse at zero frequency**, the **modulated signal** in this scheme does not contain a discrete component of the carrier frequency f_c
- Hence, the modulation process does not introduce a sinusoid at f_c, For this reason, it is called **double-sideband suppressed carrier (DSB-SC) modulation.**

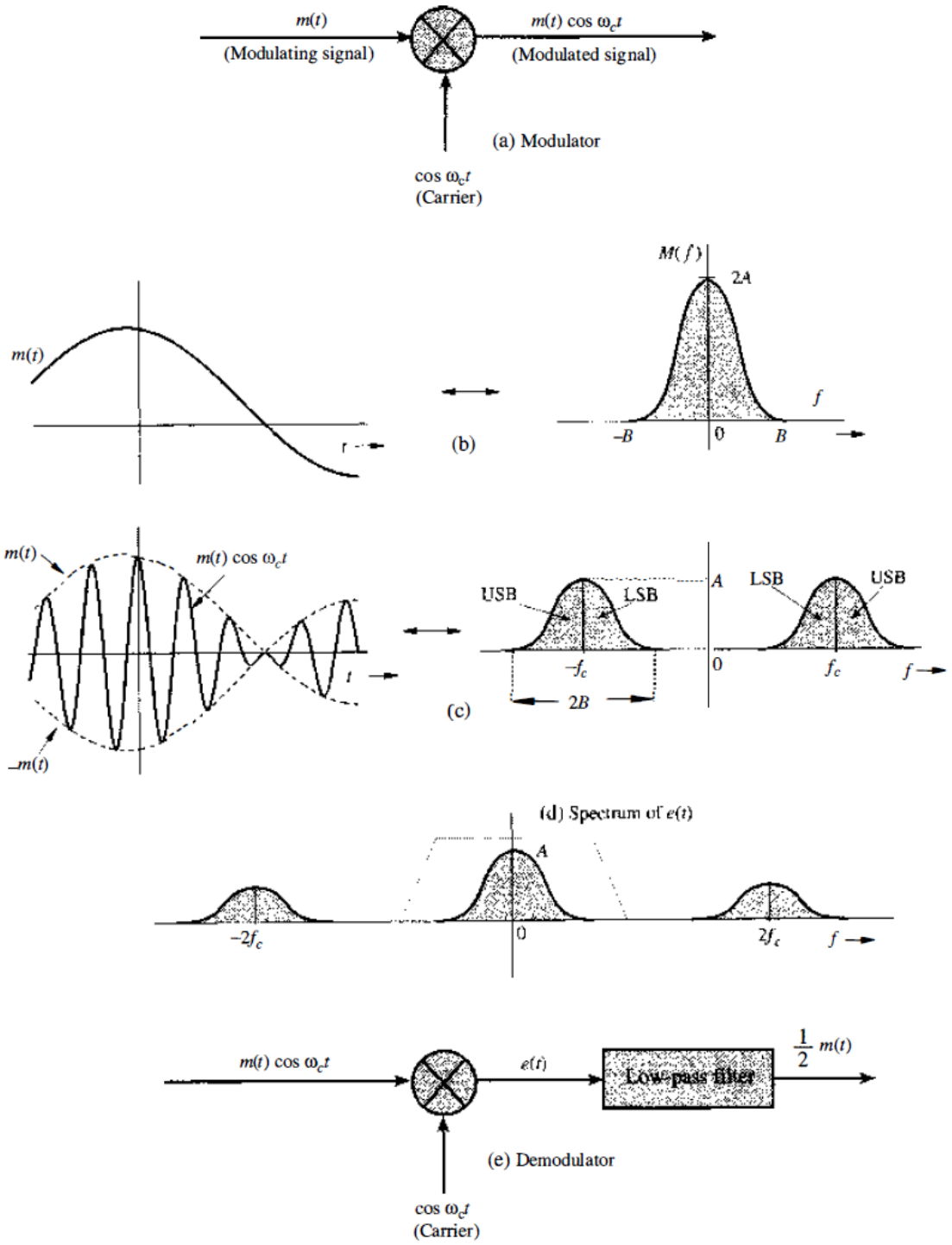


Figure 4.1. DSB-SC modulation and demodulation.

The relationship of B to f_c is of interest.

- Figure 4.1c shows that $f_c \geq B$, thus **avoiding overlap** of the modulated spectra centered at f_c and $-f_c$.

- If $f_c < B$, then the **two copies of message spectra overlap**, and **the information of *m(t)* is lost during modulation,** which makes it impossible to get back *m(t)* from the modulated signal $m(t)\cos\omega_c t$.

DEMODULATION

The **DSB-SC modulation** translates or shifts the frequency spectrum to the left and the right by f_c (i.e., at $+f_c$ and $-f_c$), as seen from Eq. (4.1).

To recover the original signal m(t) from the modulated signal, it is necessary to retranslate the spectrum to its original position.

> The **process of recovering the signal** from the modulated signal (retranslating the spectrum to its original position) is referred to as **demodulation.**

Observe that **if the modulated signal spectrum** in Fig. 4.lc is shifted to the left and to the right by f_c (and multiplied by one-half), we obtain the spectrum shown in Fig. 4.l d, **which contains the desired baseband spectrum plus** an unwanted spectrum at $\pm 2f_c$ where the latter can be suppressed by a low-pass filter.

> Thus, **demodulation**, which is almost identical to modulation, **consists of the multiplication of the incoming modulated signal** $m(t)\cos\omega_c t$ **by a carrier** $\cos\omega_c t$ followed by a low-pass filter, as shown in Fig. 4.1 e.

We can **verify this conclusion directly in the time domain** by observing that the signal *e (t)* in Fig. 4.le is

$$\begin{aligned} e(t) &= m(t)\cos^2\omega_c t \\ &= \frac{1}{2}[m(t) + m(t)\cos 2\omega_c t] \end{aligned} \tag{4.2a}$$

Therefore, the Fourier transform of the signal *e (t)* is

$$E(f) = \frac{1}{2}M(f) + \frac{1}{4}[M(f + 2f_c) + M(f - 2f_c)] \tag{4.2b}$$

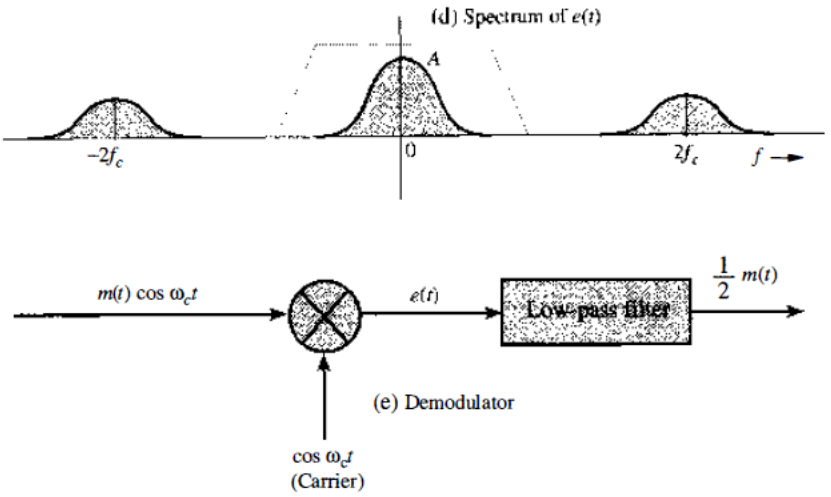


Figure 4.1. DSB-SC demodulation.

- This analysis shows that the signal e(t) consists of two components [(1/2)m(t)] and [(l/2)m(t)cos2W$_c$t], with their nonoverlapping spectra as shown in Fig.4.ld.
- The spectrum of the **second component**, being a modulated signal with carrier frequency 2f$_c$, is centered at ±2f$_c$. Hence, this component is suppressed by the low-pass filter in Fig.4.l e.
- The **desired component** (l/2)M(f), being a low-pass spectrum (centered at *f* = 0), passes through the filter unharmed, resulting in the output (1/2)m(t).
- A possible form of low pass filter characteristics is shown (under the dotted line) in Fig.4.1d. The filter leads to distortionless demodulation of the message signal m(t) from the DSB-SC signal.
- We can get rid of the inconvenient fraction 1/2 in the output by using a carrier 2 cosW$_c$t instead of cosW$_c$ t.

SYNCHRONOUS DETECTION, OR COHERENT DETECTION

The above method of recovering the baseband signal is called **synchronous detection**, **or coherent detection**, where **we use a carrier of exactly the same frequency (and phase) as the carrier used for modulation**.

Thus, for **demodulation**, **we need to generate a local carrier at the receiver in frequency and phase coherence (synchronism) with the carrier used at the modulator.**

Example 4.1 For a baseband signal

$$m(t) = \cos\ \omega_m t = \cos\ 2\pi f_m t,$$

find the DSB-SC signal, and sketch its spectrum. Identify the USB and LSB. Verify that the DSB-SC modulated signal can be demodulated by the demodulator in Fig. 4.1e.

Solution

The case in this example is referred to as *tone modulation* because the modulating signal is a pure sinusoid, or tone, $\cos\ \omega_m t$. To clarify the basic concepts of DSB-SC modulation, we shall work this problem in the frequency domain as well as the time domain. In the frequency domain approach, we work with the signal spectra. The spectrum of the baseband signal $m(t) = \cos\ \omega_m t$ is given by

$$\begin{aligned} M(f) &= \frac{1}{2}[\delta(f - f_m) + \delta(f + f_m)] \\ &= \pi[\delta(\omega - \omega_m) + \delta(\omega + \omega_m)] \end{aligned}$$

The message spectrum consists of two impulses located at $\pm f_m$, as shown in Fig. 4.2a. The DSB-SC (modulated) spectrum, as seen from Eq. (4.1), is the baseband spectrum in Fig. 4.2a shifted to the right and the left by f_c (times one-half), as shown in Fig. 4.2b. This spectrum consists of impulses at angular frequencies $\pm(f_c - f_m)$ and $\pm(f_c + f_m)$. The spectrum beyond f_c is the USB, and the one below f_c is the LSB. Observe that the DSB-SC spectrum does not have the component of the carrier frequency f_c. This is why it is called *suppressed carrier*.

In the time domain approach, we work directly with signals in the time domain. For the baseband signal $m(t) = \cos\ \omega_m t$, the DSB-SC signal $\varphi_{\text{DSB-SC}}(t)$ is

$$\begin{aligned} \varphi_{\text{DSB-SC}}(t) &= m(t)\cos\ \omega_c t \\ &= \cos\ \omega_m t \cos\ \omega_c t \\ &= \frac{1}{2}[\cos(\omega_c + \omega_m)t + \cos(\omega_c - \omega_m)t] \end{aligned}$$

This shows that when the baseband (message) signal is a single sinusoid of frequency f_m, the modulated signal consists of two sinusoids: the component of frequency $f_c + f_m$ (the USB) and the component of frequency $f_c - f_m$ (the LSB). Figure 4.2b shows precisely the spectrum of $\varphi_{\text{DSB-SC}}(t)$. Thus, each component of frequency f_m in the modulating signal turns into two components of frequencies $f_c + f_m$ and $f_c - f_m$ in the modulated signal. Note the curious fact that there is no component of the carrier frequency f_c on the right-hand side of the preceding equation. As mentioned, this is why it is called double-sideband suppressed carrier (DSB-SC) modulation.

We now verify that the modulated signal $\varphi_{\text{DSB-SC}}(t) = \cos\ \omega_m t \cos\ \omega_c t$, when applied to the input of the demodulator in Fig. 4.1e, yields the output proportional to the desired baseband signal $\cos\ \omega_m t$. The signal $e(t)$ in Fig. 4.1e is given by

$$\begin{aligned} e(t) &= \cos\ \omega_m t \cos^2 \omega_c t \\ &= \frac{1}{2}\cos\ \omega_m t\,(1 + \cos\ 2\omega_c t) \end{aligned}$$

The spectrum of the term $\cos\ \omega_m t \cos\ 2\omega_c t$ is centered at $2\omega_c$ and will be suppressed by the low-pass filter, yielding $\frac{1}{2}\cos\ \omega_m t$ as the output. We can also derive this result in the frequency domain. Demodulation causes the spectrum in Fig. 4.2b to shift left and right by ω_c (and to be multiplied by one-half). This results in the spectrum shown in Fig. 4.2c. The low-pass filter suppresses the spectrum centered at $\pm 2\omega_c$, yielding the spectrum $\frac{1}{2}M(f)$.

$$e(t) = \cos\ \omega_m t \cos^2 \omega_c t$$
$$= \frac{1}{2} \cos\ \omega_m t\ (1 + \cos\ 2\omega_c t)$$

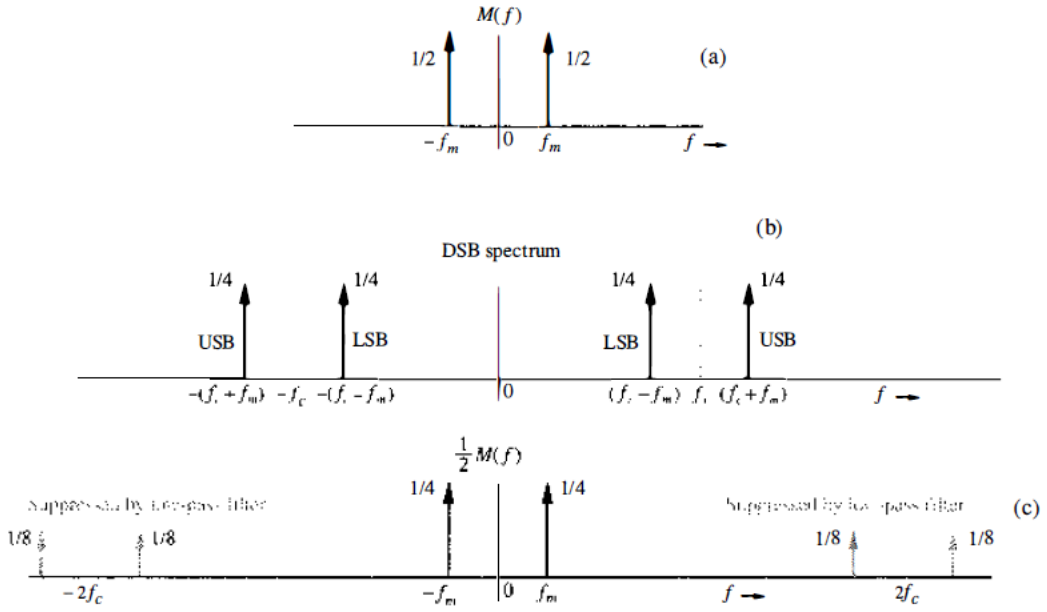


Figure 4.2. Example of DSB-SC modulation.

AMPLITUDE MODULATION (AM)

In the last section, we began our discussion of amplitude modulation by introducing the DSBSC amplitude modulation because it is **easy to understand and to analyze in both the time and frequency domains**.

However, **analytical simplicity does not always equate to simplicity in practical implementation**.

The **(coherent) demodulation of a DSB-SC** signal requires **the receiver to possess a carrier signal** that is synchronized with the incoming carrier. This requirement is not easy to achieve in practice.

- Because the modulated signal may have traveled hundreds of miles and could even suffer from some unknown frequency shift, the **bandpass received signal** in fact has the form of

$$r(t) = A_c m(t - t_0) \cos\ [(\omega_c + \Delta\omega)(t - t_0)] = A_c m(t - t_0) \cos\ [(\omega_c + \Delta\omega)t - \theta_d)]$$

in which $\Delta\omega$ represents the Doppler effect while

$$\theta_d = (\omega_c + \Delta\omega)t_d$$

comes from the unknown delay t_0.

To utilize the coherent demodulator, the receiver must be sophisticated enough to generate a local oscillator $\cos\left[\left(\omega_c+\Delta\omega\right)t-\theta_d\right]$ purely from the received signal *r(t)* .

- **Such a receiver** would be harder to implement and could be quite costly.
- This cost is particularly to be avoided in broadcasting systems, which have many receivers for every transmitter.

The **alternative to a coherent demodulator** is for the transmitter to send a carrier $A\cos\omega_c t$ [along with the modulated signal $m(t)\cos\omega_c t$] **so that there is no need to generate a carrier at the receiver.**

In this case the transmitter needs to transmit at a **much higher power level**, which increases its cost as a trade-off.

In point-to-point communications, where there is one transmitter for every receiver, substantial complexity in the receiver system can be justified, **provided its cost is offset by a less expensive transmitter**.

On the other hand, **for a broadcast system** with a huge number of receivers for each transmitter, it is more economical to have one expensive high-power transmitter and simpler, less expensive receivers because any cost saving at the receiver is **multiplied by the number of receiver units**.

- For this reason, broadcasting systems tend to favor the trade-off by migrating cost from the (many) receivers to the (fewer) transmitters.
- **The second option** of transmitting a carrier along with the modulated signal is the obvious choice in broadcasting because of its desirable trade-offs.
- **This leads to the so-called AM** (amplitude modulation), in which the transmitted signal $\varphi_{AM}(t)$ is given by

$$\varphi_{AM}(t) = A\cos\ \omega_c t + m(t)\cos\ \omega_c t \quad (4.8a)$$

$$= [A + m(t)]\cos\ \omega_c t \quad (4.8b)$$

The spectrum of $\varphi_{AM}(t)$ is basically the same as that of $\varphi_{DSB-SC}(t) = m(t)\cos\ \omega_c t$ except for the two additional impulses at $\pm f_c$,

$$\varphi_{AM}(t) \iff \frac{1}{2}[M(f+f_c) + M(f-f_c)] + \frac{A}{2}[\delta(f+f_c) + \delta(f-f_c)] \quad (4.8c)$$

Upon comparing $\varphi_{AM}(t)$ with $\varphi_{DSB-SC}(t) = m(t)\cos\ \omega_c t$, it is clear that the AM signal is identical to the DSB-SC signal with $A + m(t)$ as the modulating signal [instead of $m(t)$]. The value of A is always chosen to be positive. Therefore, to sketch $\varphi_{AM}(t)$, we sketch the envelope $|A + m(t)|$ and its mirror image $-|A + m(t)|$ and fill in between with the sinusoid of the carrier frequency f_c. The size of A affects the time domain envelope of the modulated signal.

Two cases are considered in Fig. 4.8.

1. **In the first case, A is large enough** that $A + m(t) \geq 0$ is always **nonnegative**.
2. **In the second case, *A* is not large enough to satisfy this condition**.

In the first case, the **envelope has the same shape as m(t)** (although riding on a dc of magnitude A).

In the second case, the **envelope shape differs from the shape of m(t)** because the negative part of $A + m(t)$ is rectified.

This means we can detect the desired signal *m(t)* by detecting the envelope in the first case when $A + m(t) > 0$. **Such detection is not possible in the second case**.

We shall see that **envelope detection is an extremely simple and inexpensive operation**, which **does not require generation of a local carrier for the demodulation.**

But as seen earlier, the AM envelope has the information about *m(t)* only if the AM signal $[A + m(t)]\cos\omega_c t$ satisfies the condition $A + m(t) > 0$ for all t.

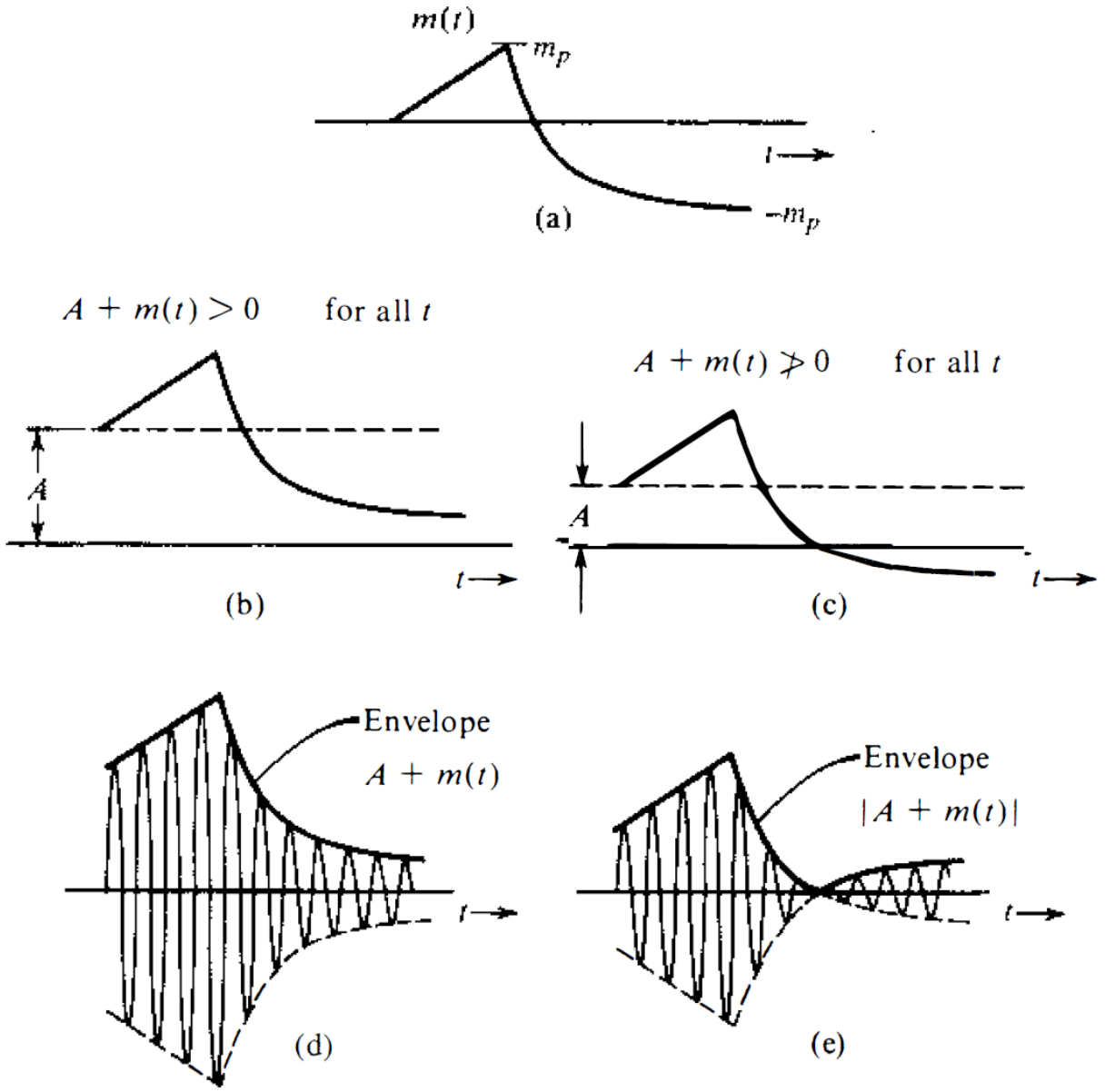


Figure 4.8. AM signal and its envelope.

Envelope Detection

Let us now be more precise about the definition of "envelope." Consider a signal $E(t)\cos\omega_c t$. If $E(t)$ varies slowly in comparison with the sinusoidal carrier $\cos\omega_c t$, then the **envelope** of $E(t)\cos\omega_c t$ is $|E(t)|$. This means [see Eq. (4.8b)] that if and only if $A + m(t) \geq 0$ for all t, the envelope of $\varphi_{\text{AM}}(t)$ is

$$|A + m(t)| = A + m(t)$$

In other words, for envelope detection to properly detect $m(t)$, two conditions must be met:

(a) $f_c \gg$ bandwidth of $m(t)$

(b) $A + m(t) \geq 0$

This conclusion is readily verified from Fig. 4.8d and e. In Fig. 4.8d, where $A + m(t) \geq 0$, $A + m(t)$ is indeed the envelope, and $m(t)$ can be recovered from this envelope. In Fig. 4.8e, where $A + m(t)$ is not always positive, the envelope $|A + m(t)|$ is rectified from $A + m(t)$, and $m(t)$ cannot be recovered from the envelope. Consequently, demodulation of $\varphi_{AM}(t)$ in Fig. 4.8d amounts to simple envelope detection. Thus, the **condition for envelope detection** of an AM signal is

$$A + m(t) \geq 0 \qquad \text{for all } t \tag{4.9a}$$

Special cases

If $m(t) \geq 0$ for all t, then $A = 0$ already satisfies condition (4.9a). In this case there is no need to add any carrier because the envelope of the DSB-SC signal $m(t) \cos \omega_c t$ is $m(t)$ and such a DSB-SC signal can be detected by envelope detection. In the following discussion we assume that $m(t) \not\geq 0$ for all t; that is, $m(t)$ can be negative over some range of t.

Message Signals $m(t)$ with Zero Offset: Let $\pm m_p$ be the maximum and the minimum values of $m(t)$, respectively (see Fig. 4.8). This means that $m(t) \geq -m_p$. Hence, the condition of envelope detection (4.9a) is equivalent to

$$A \geq -m_{\min} \tag{4.9b}$$

Thus, the minimum carrier amplitude required for the viability of envelope detection is m_p. This is quite clear from Fig. 4.8. We define the modulation index μ as

$$\mu = \frac{m_p}{A} \tag{4.10a}$$

For envelope detection to be distortionless, the condition is $A \geq m_p$. Hence, it follows that

$$0 \leq \mu \leq 1 \tag{4.10b}$$

is the required condition for the distortionless demodulation of AM by an envelope detector.

When $A < m_p$, Eq. (4.10a) shows that $\mu > 1$ (overmodulation). In this case, the option of envelope detection is no longer viable. We then need to use synchronous demodulation.

Note that **synchronous demodulation can be used for any value of μ,**

- since the demodulator will recover signal $A + m(t)$.
- Only **an additional de block** is needed **to remove** the DC voltage A.

The **envelope detector**, which is considerably simpler and less expensive than the synchronous detector, can be used **only for** $\mu \leq 1$.

Message Signals $m(t)$ with Nonzero Offset: On rare occasions, the message signal $m(t)$ will have a nonzero offset such that its maximum $m_{\max}$ and its minimum $m_{\min}$ are not symmetric, that is,

$$m_{\min} \neq -m_{\max}$$

In this case, it can be recognized that any offset to the envelope does not change the shape of the envelope detector output. In fact, one should note that constant offset does not carry any fresh information.

In this case, envelope detection would still remain distortionless if

$$0 \leq \mu \leq 1 \tag{4.11a}$$

with a modified modulation index definition of

$$\mu = \frac{m_{\max} - m_{\min}}{2A + m_{\max} + m_{\min}} \tag{4.11b}$$

Example 4.4 Sketch $\varphi_{AM}(t)$ for modulation indices of $\mu = 0.5$ and $\mu = 1$, when $m(t) = b \cos \omega_m t$. This case is referred to as **tone modulation** because the modulating signal is a pure sinusoid (or tone).

Solution

In this case, $m_{\max} = b$ and $m_{\min} = -b$. Hence the modulation index according to Eq. (4.10a) is

$$\mu = \frac{b - (-b)}{2A + b + (-b)} = \frac{b}{A}$$

Hence, $b = \mu A$ and

$$m(t) = b \cos \omega_m t = \mu A \cos \omega_m t$$

Therefore,

$$\varphi_{AM}(t) = [A + m(t)] \cos \omega_c t = A[1 + \mu \cos \omega_m t] \cos \omega_c t$$

Figure 4.9 shows the modulated signals corresponding to $\mu = 0.5$ and $\mu = 1$, respectively.

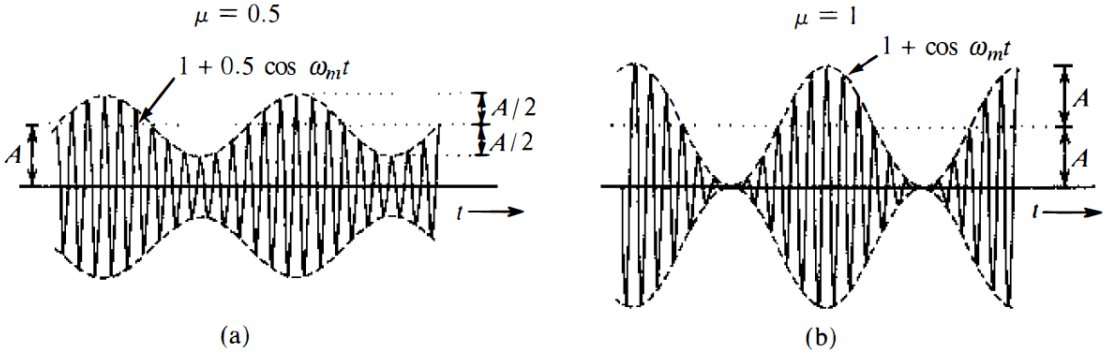


Figure 4.9. Tone-modulated AM. (a) μ = 0.5. (b) μ = 1.

Sideband and Carrier Power

The advantage of envelope detection in AM comes at a price. In AM, the carrier term does not carry any information, and hence, the carrier power is wasteful from this point of view:

$$\varphi_{\mathrm{AM}}(t) = \underbrace{A\cos\ \omega_c t}_{\text{carrier}} + \underbrace{m(t)\cos\ \omega_c t}_{\text{sidebands}}$$

The carrier power P_c is the mean square value of $A\cos\ \omega_c t$, which is $A^2/2$. The sideband power P_s is the power of $m(t)\cos\ \omega_c t$, which is 0.5 $\overline{m^2(t)}$ [see Eq. (3.93)]. Hence,

$$P_c = \frac{A^2}{2} \qquad \text{and} \qquad P_s = \frac{1}{2}\overline{m^2(t)}$$

The useful message information resides in the sideband power, whereas the carrier power is the used for convenience in modulation and demodulation. The total power is the sum of the carrier (wasted) power and the sideband (useful) power. Hence, η, the power efficiency, is

$$\eta = \frac{\text{useful power}}{\text{total power}} = \frac{P_s}{P_c + P_s} = \frac{\overline{m^2(t)}}{A^2 + \overline{m^2(t)}}100\%$$

For the special case of tone modulation,

$$m(t) = \mu A\cos\ \omega_m t \qquad \text{and} \qquad \overline{m^2(t)} = \frac{(\mu A)^2}{2}$$

Hence

$$\eta = \frac{\mu^2}{2+\mu^2}100\%$$

with the condition that $0 \le \mu \le 1$. It can be seen that η increases monotonically with μ, and $\eta_{\max}$ occurs at $\mu = 1$, for which

$$\eta_{\max} = 33\%$$

Thus, for tone modulation, under the best conditions ($\mu = 1$), only one-third of the transmitted power is used for carrying messages. For practical signals, the efficiency is even worse—on the order of 25% or lower—compared with the DSB-SC case. The best condition implies $\mu = 1$.

Example 4.5 Determine η and the percentage of the total power carried by the sidebands of the AM wave for tone modulation when $\mu = 0.5$ and when $\mu = 0.3$.

For $\mu = 0.5$,

$$\eta = \frac{\mu^2}{2+\mu^2}100\% = \frac{(0.5)^2}{2+(0.5)^2}100\% = 11.11\%$$

Hence, only about 11% of the total power is in the sidebands. For $\mu = 0.3$,

$$\eta = \frac{(0.3)^2}{2+(0.3)^2}100\% = 4.3\%$$

Hence, only 4.3% of the total power is in the sidebands that contain the message signal.

Generation of AM Signals

In principle, **the generation of AM signals** is **identical to that of the DSB-SC modulations** discussed **earlier** except that an additional carrier component $A\cos\omega_c t$ needs to be added to the DSB-SC signal.

Demodulation of AM Signals

Like DSB-SC signals, **the AM signal can be demodulated coherently by a locally generated carrier**.

- **Coherent, or synchronous, demodulation** of AM, however, defeats the purpose of AM because it does not take advantage of the additional carrier component $A\cos\omega_c t$.
- As we have seen earlier, for $\mu \le 1$, the envelope of the AM signal follows the message signal *m(t)*.
- Hence, we shall consider two **noncoherent methods** of AM demodulation under the condition of $0 \le \mu \le 1$:
  - rectifier detection and
  - envelope detection.

Rectifier Detector

If an AM signal is applied to a diode and a resistor circuit (Fig. 4.10), the negative part of the AM wave will be removed.

- The output across the resistor is a half-wave rectified version of the AM signal.

Visually, the diode acts like a pair of scissors by cutting off any negative half-cycle of the modulated sinusoid.

- In essence, **at the rectifier output, the AM signal is multiplied by *w*(t)**.

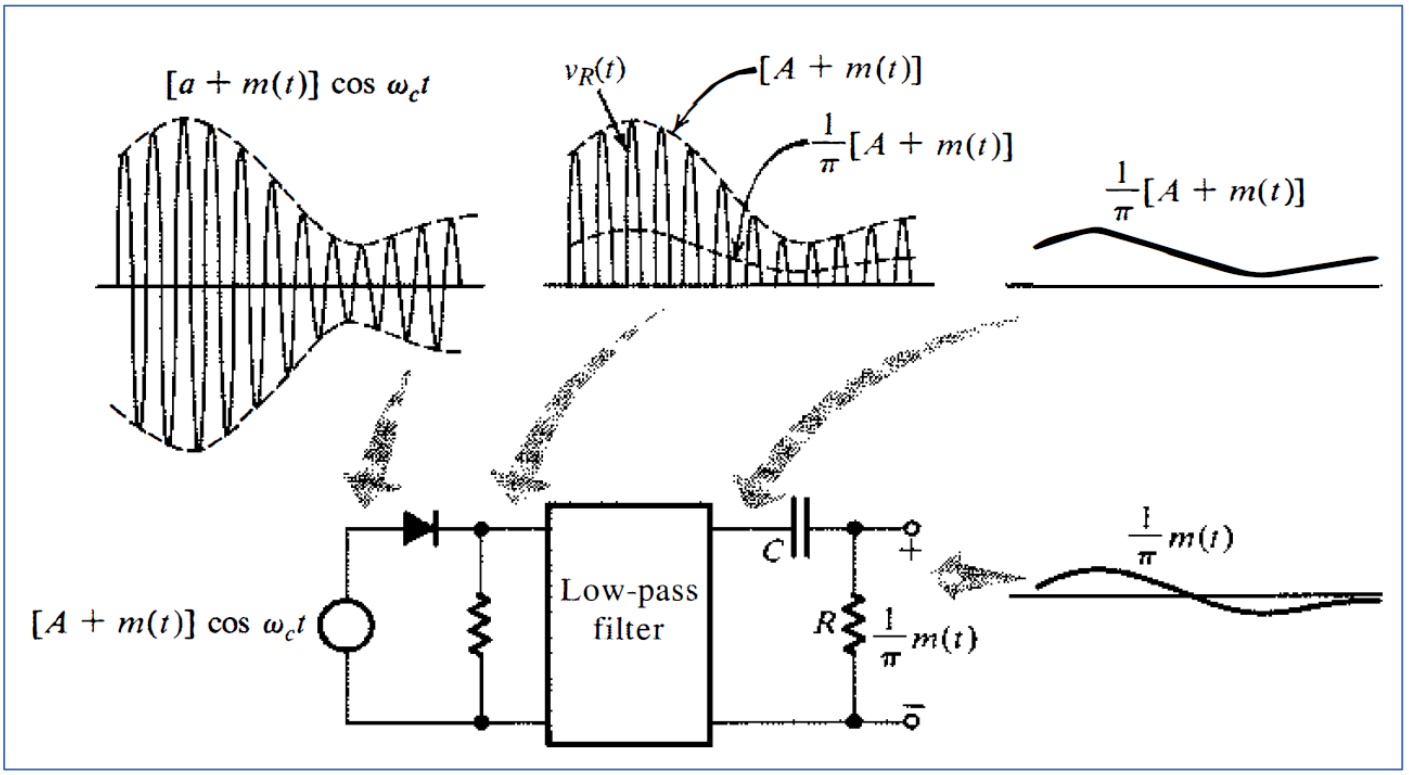


Figure 4.10. Rectifier detector for AM.

Hence, the half-wave-rectified output $V_R(t)$ is

$$v_R(t) = \{[A + m(t)]\cos\ \omega_c t\}\, w(t) \tag{4.12}$$

$$= [A + m(t)]\cos\ \omega_c t \left[\frac{1}{2} + \frac{2}{\pi}\left(\cos\ \omega_c t - \frac{1}{3}\cos\ 3\omega_c t + \frac{1}{5}\cos\ 5\omega_c t - \cdots\right)\right] \tag{4.13}$$

$$= \frac{1}{\pi}[A + m(t)] + \text{other terms of higher frequencies} \tag{4.14}$$

- When $V_R(t)$ is **applied to a low-pass filter of cutoff *B* Hz**, the output is $[A + m(t)]/n$, and all the other terms in V_R of frequencies higher than B Hz are suppressed.

- The dc term A/n may be blocked by a capacitor (Fig. 4.10) to give the desired output $m(t)/n$.
- The output can be doubled by using a full-wave rectifier.
- It is interesting to note that **because of the multiplication with *w(t)*,** rectifier detection is in **effect synchronous detection performed without using a local carrier.**

Envelope Detector

The output of an envelope detector **follows** the envelope of the modulated signal.

The **simple circuit** shown in **Fig. 4.lla functions as an envelope detector**.

- On the **positive cycle** of the input signal, the input grows and may exceed the charged voltage on the capacity $v_C(t)$, **turning on the diode** and **allowing the capacitor C to charge up to the peak voltage** of the input signal cycle.
- **As the input signal falls** below this peak value, **it falls quickly below the capacitor voltage** (**which is very nearly the peak voltage**), thus **causing the diode to open**.
- The **capacitor now discharges** through the resistor R at **a slow rate (with a time constant *RC*)**.

During the next positive cycle, **the same drama repeats**.

- **As the input signal rises above the capacitor voltage**, **the diode conducts again.**
- **The capacitor again charges to the peak value of this (new) cycle.**
- The **capacitor discharges slowly** during the cutoff period.

During each positive cycle, **the capacitor charges up to the peak voltage of the input signal and then decays slowly until the next positive cycle** as shown in Fig. 4.11 b.

The **output voltage $v_C(t)$**, thus, **closely follows the (rising) envelope** of the input AM signal.

Equally important, **the slow capacity discharge** via the resistor R **allows the capacity voltage** to follow a declining envelope.

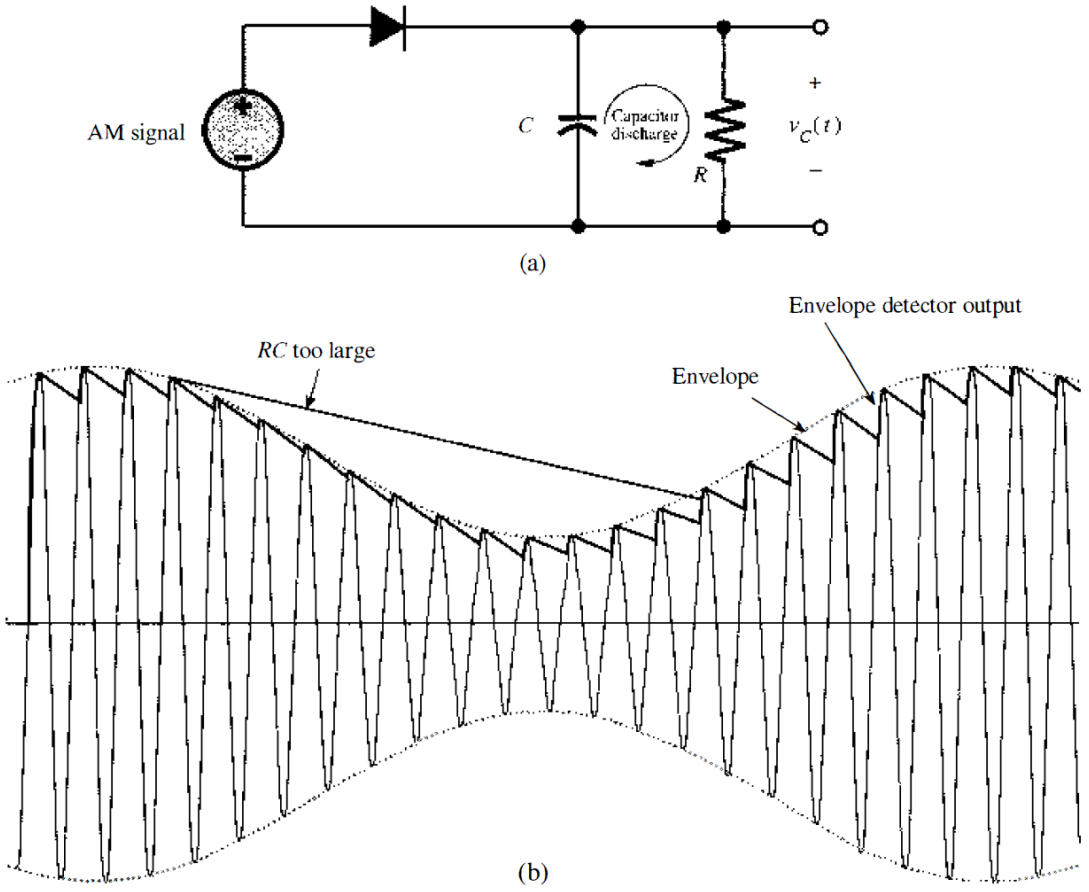


Figure 4.11. Envelope detector for AM.

Capacitor discharge between positive peaks causes a ripple signal of frequency ω_c in the output.

- This **ripple can be reduced** by choosing a **larger** time constant RC so that the capacitor discharges very little between the positive peaks ($RC \gg 1/\omega_c$).
- Picking ***RC* too large**, however, would make it impossible for the capacitor voltage to follow a fast-declining envelope (see **Fig.4.l lb**).
- Because **the maximum rate of AM envelope decline** is dominated by the bandwidth B of the message signal m(t), **the design criterion of *RC* should be**

$$1/\omega_c \ll RC < 1/(2\pi B) \qquad \text{or} \qquad 2\pi B < \frac{1}{RC} \ll \omega_c$$

The **envelope detector output** is $v_C(t) = A + m(t)$ with a **ripple of frequency** ω_c.

- The **dc term *A*** can be blocked out by **a capacitor** or a **simple *RC* high-pass filter**.
- The **ripple may be reduced further** by another (low-pass) RC filter.

Bandwidth-Efficient Amplitude Modulations

From Figure 4.12, the **DSB spectrum** (including suppressed carrier and AM) has **two sidebands**:

- the **upper sideband (USB)** and the **lower sideband (LSB), each containing the complete information of the baseband signal *m(t)*.**
- As a result, for a baseband signal ***m(t)*** with bandwidth *B* Hz, **DSB modulations require twice the radio-frequency bandwidth** to transmit.

To improve the spectral efficiency of amplitude modulation, there exist **two basic schemes to either utilize or remove the 100% spectral redundancy**:

- **Single-sideband (SSB)** modulation, which **removes** either the **LSB** or the **USB** that uses only a bandwidth of *B* Hz for one message signal *m(t)*.
- **Quadrature amplitude modulation (QAM)**, which **utilizes** spectral redundancy by sending two messages over the same bandwidth of *2B* Hz.

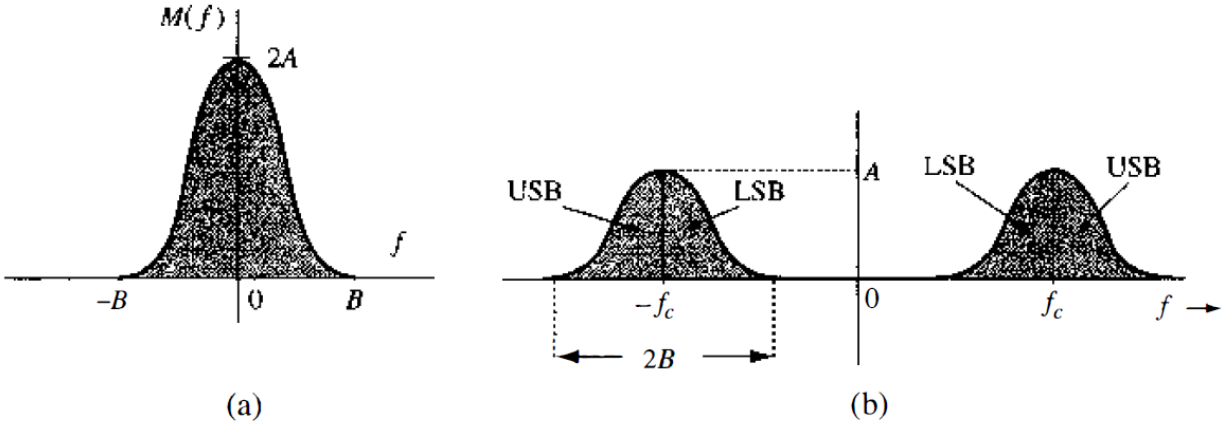


Figure 4.12. (a) Original message spectrum. (b) The redundant bandwidth consumption in DSB modulations.

Single Sideband (SSB) Amplitude Modulation

As shown in Fig. 4.13, **either the LSB or the USB can be suppressed** from the DSB signal **via bandpass filtering**.

Such a scheme in which **only one sideband is transmitted** is known as **single-sideband (SSB) transmission,** and **requires only one-half the bandwidth** of the DSB signal.

(Coherent Demodulation)

An SSB signal can be coherently (synchronously) demodulated just like DSB-SC signals.

- For example, **multiplication of a USB signal** (Fig. 4.13c) by $\cos\omega_c t$ shifts its spectrum to the left and right by ω_c, yielding the spectrum in Fig. 4.13e.
- **Low-pass filtering** of this signal yields the desired baseband signal.
- The case is **similar with LSB** signals.

Since the **demodulation of SSB** signals is identical to that of DSB-SC signals, the **transmitters can now utilize only half the DSB-SC signal bandwidth** without any additional cost to the receivers.

Since no additional carrier accompanies the modulated SSB signal, the resulting modulator outputs are known as **suppressed carrier signals (SSB-SC).**

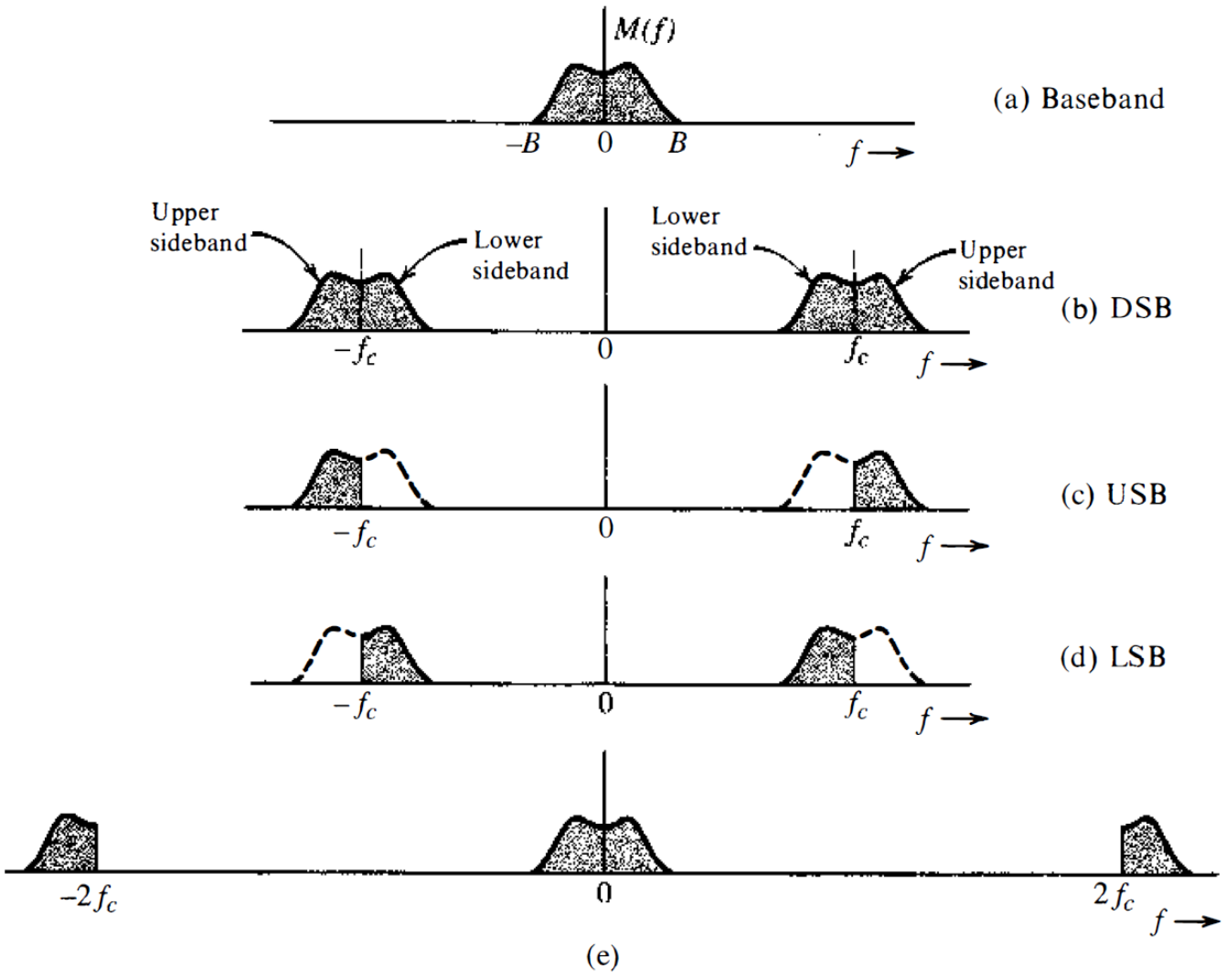


Figure 4.13. SSB spectra from suppressing one DSB sideband.

VESTIGIAL SIDEBAND (VSB) AMPLITUDE MODULATIONS

As discussed earlier, the generation of **DSB signals** is much simpler, but it **requires twice the signal bandwidth**. Also, it is **difficult to generate exact SSB** signals.

Vestigial sideband (VSB) modulation, also called the **asymmetric sideband system**, is a **compromise between DSB and SSB**.

- It inherits the advantages of DSB and SSB but avoids their disadvantages **at a small cost**.
- VSB signals are relatively **easy to generate**, and, at the same time, their **bandwidth is only a little (typically 25%) greater than that of SSB** signals.

In VSB, instead of rejecting one sideband completely (as in SSB), **a gradual cutoff of one sideband** as shown in Fig. 4.20d, is accepted [1].

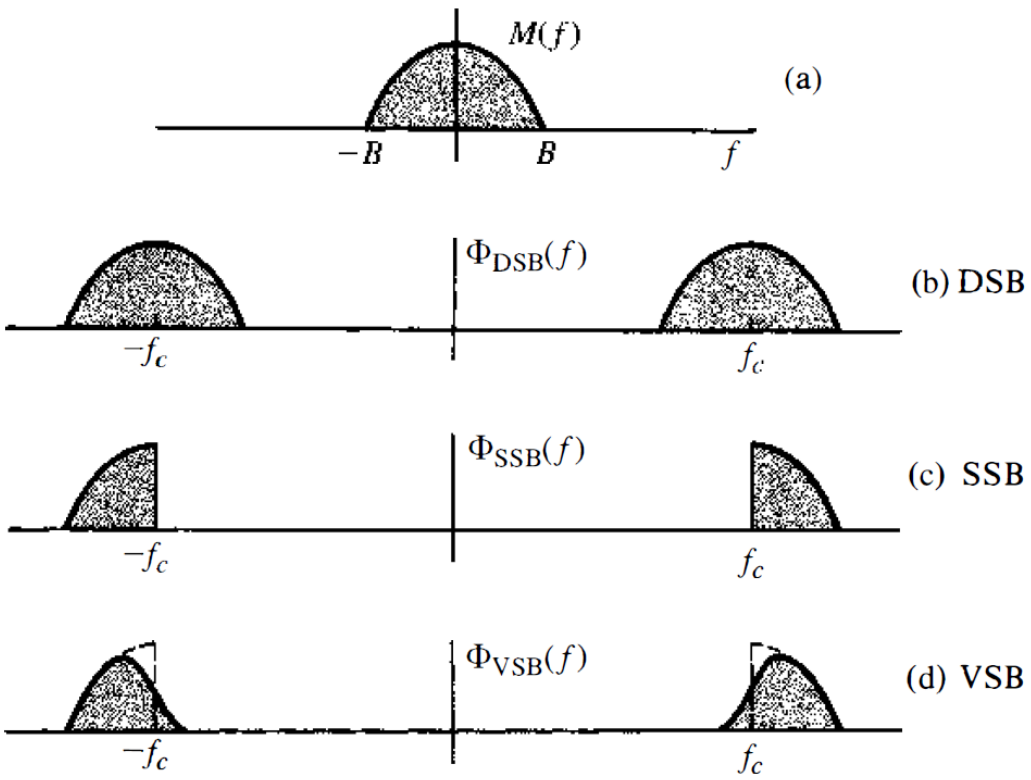


Figure 4.20. Spectra of the modulating signal and corresponding DSB, SSB, and VSB signals [1].

More specifically [2],

In *vestigial sideband (VSB) modulation*, one of the sidebands is partially suppressed and a vestige of the other sideband is transmitted to compensate for that suppression. A popular method for generating a VSB-modulated wave is to use the *frequency discrimination method*. First, we generate a DSB-SC modulated wave and then pass it through a band-pass filter, as shown in Figure 2.12; it is the special design of the band-pass filter that distinguishes VSB modulation from SSB modulation. Assuming that a vestige of the lower sideband is transmitted, the frequency response $H(f)$ of the band-pass filter takes the form shown in Figure 2.13. To simplify matters, only the response for positive frequencies is shown here. This frequency response is normalized, so that at the carrier frequency f_c we have $|H(f_c)| = 1/2$. The important feature to note from Figure 2.13 is that the cutoff portion of the frequency response around the carrier frequency f_c exhibits *odd symmetry*. Note also that outside the frequency band of interest (i.e., $|f| > f_c + W$), the frequency response $H(f)$ may have an arbitrary specification. Accordingly, the transmission bandwidth of VSB modulation is

$$B_T = W + f_v$$

where W is the message bandwidth, and f_v is the width of the vestigial sideband.

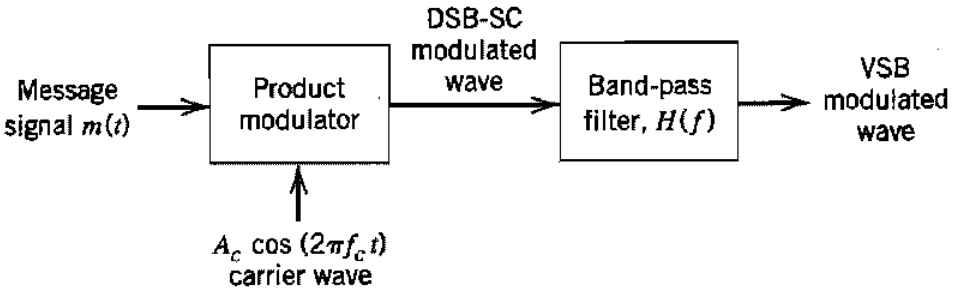


Figure 2.12. Filtering scheme for the generation of VSB modulated wave [2].

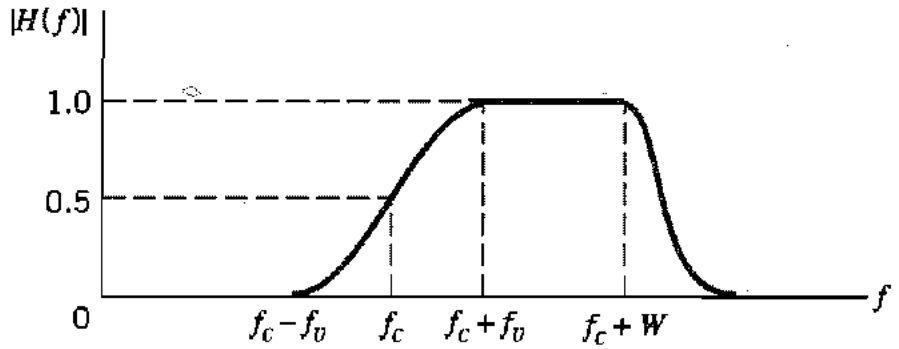


Figure 2.13. Magnitude response of VSB filter [2].

The baseband signal can be **recovered exactly by a synchronous detector** in conjunction with an appropriate filter at the receiver.

References

[1] B. P. Lathi and Z. Ding, "Modern Digital and Analog Communication System," 4th edition, oxford university press, 2010.
[2] Simon S. Haykin and Micheal Mohar "Communication Systems," 5th edition, Wiley, 2009.

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